# Design and Calculation of an LC-Based Dual Active Bridge (DAB) Converter

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## Contents

1 Introduction 2

2 Design Specifications 2

3 Transformer Turns Ratio 2

4 HV-Side Resonant Capacitor Calculation 2

5 Verification of Resonant Frequency 3

6 Characteristic Impedance 3

7 Ideal Input and Output Currents 4

8 Equivalent Resistive Load 4

9 HV-Side Tank Referred to the LV Side 4

10 Simulation Notes 5

# 1 Introduction

This document presents the preliminary design calculations for an LC-based Dual Active Bridge (DAB) converter. The resonant inductor and capacitor are placed on the high-voltage (HV) side of the high-frequency transformer.

The target resonant frequency is 30 kHz, with an LV-side voltage of 28 V and an HV-side voltage of 504 V.

# 2 Design Specifications

Table 1: Converter design specifications

| **Parameter** | **Symbol** | **Value** |
|---|---|---|
| LV-side voltage | $V_{\mathrm{LV}}$ | 28 V |
| HV-side voltage | $V_{\mathrm{HV}}$ | 504 V |
| Rated power | $P$ | 3 kW |
| Switching frequency | $f_s$ | 30 kHz |
| Target resonant frequency | $f_r$ | 30 kHz |
| Transformer turns ratio | $n = N_{\mathrm{HV}}/N_{\mathrm{LV}}$ | 18 |
| HV-side resonant inductance | $L_{cr}$ | 40 µH |
| HV-side resonant capacitance | $C_{cr}$ | To be calculated |

# 3 Transformer Turns Ratio

The transformer turns ratio is defined as

$$n = \frac{N_{\mathrm{HV}}}{N_{\mathrm{LV}}} = 18. \tag{1}$$

The HV-side voltage referred to the LV side is

$$V'_{\mathrm{HV}} = \frac{V_{\mathrm{HV}}}{n}. \tag{2}$$

Substituting the values,

$$V'_{\mathrm{HV}} = \frac{504}{18} = \boxed{28\,\mathrm{V}}. \tag{3}$$

Thus, the ideal transformer voltage ratio matches the specified LV-side voltage.

# 4 HV-Side Resonant Capacitor Calculation

For a series LC resonant tank, the resonant frequency is

$$f_r = \frac{1}{2\pi\sqrt{L_{cr}C_{cr}}}. \tag{4}$$

Rearranging the equation to calculate the capacitance,

$$C_{cr} = \frac{1}{(2\pi f_r)^2 L_{cr}}. \tag{5}$$

Given:

$$f_r = 30\,\text{kHz}, \tag{6}$$
$$L_{cr} = 40\,\mu\text{H}. \tag{7}$$

Substituting the values,

$$C_{cr} = \frac{1}{(2\pi \times 30000)^2 (40 \times 10^{-6})}. \tag{8}$$

Therefore,

$$\boxed{C_{cr} \approx 0.704\,\mu\text{F}}. \tag{9}$$

Hence, the selected HV-side resonant tank parameters are

$$\boxed{L_{cr} = 40\,\mu\text{H}}, \tag{10}$$
$$\boxed{C_{cr} = 0.704\,\mu\text{F}}. \tag{11}$$

# 5 Verification of Resonant Frequency

The resonant frequency is verified using the calculated inductance and capacitance:

$$f_r = \frac{1}{2\pi\sqrt{(40 \times 10^{-6})(0.704 \times 10^{-6})}}. \tag{12}$$

The resulting resonant frequency is approximately

$$\boxed{f_r \approx 30\,\text{kHz}}. \tag{13}$$

The calculated resonant frequency matches the target switching frequency in the ideal LC model.

# 6 Characteristic Impedance

The characteristic impedance of the series LC tank is

$$Z_r = \sqrt{\frac{L_{cr}}{C_{cr}}}. \tag{14}$$

Substituting the values,

$$Z_r = \sqrt{\frac{40 \times 10^{-6}}{0.704 \times 10^{-6}}}. \tag{15}$$

Therefore,

$$\boxed{Z_r \approx 7.54\,\Omega}. \tag{16}$$

# 7 Ideal Input and Output Currents

Assuming an ideal converter with no losses, the input current at rated power is

$$I_{\mathrm{LV}} = \frac{P}{V_{\mathrm{LV}}}. \tag{17}$$

$$I_{\mathrm{LV}} = \frac{3000}{28} \approx \boxed{107.14\,\mathrm{A}}. \tag{18}$$

The output current at rated power is

$$I_{\mathrm{HV}} = \frac{P}{V_{\mathrm{HV}}}. \tag{19}$$

$$I_{\mathrm{HV}} = \frac{3000}{504} \approx \boxed{5.95\,\mathrm{A}}. \tag{20}$$

These values are ideal DC input and output currents. They are not the peak or RMS currents of the resonant tank.

# 8 Equivalent Resistive Load

For a resistive load delivering 3 kW at 504 V, the equivalent load resistance is

$$R_{\mathrm{load}} = \frac{V_{\mathrm{HV}}^2}{P}. \tag{21}$$

Substituting the values,

$$R_{\mathrm{load}} = \frac{504^2}{3000}. \tag{22}$$

Therefore,

$$\boxed{R_{\mathrm{load}} \approx 84.67\,\Omega}. \tag{23}$$

# 9 HV-Side Tank Referred to the LV Side

For reference and analytical comparison, the HV-side impedance can be referred to the LV side using the ideal transformer turns ratio $n = 18$.

The equivalent inductance is

$$L'_{cr} = \frac{L_{cr}}{n^2}. \tag{24}$$

$$L'_{cr} = \frac{40}{18^2} \approx \boxed{0.1235\,\mu\mathrm{H}}. \tag{25}$$

The equivalent capacitance is

$$C'_{cr} = n^2 C_{cr}. \tag{26}$$

$$C'_{cr} = 18^2 \times 0.704 \approx \boxed{228.1\,\mu\mathrm{F}}. \tag{27}$$

The equivalent characteristic impedance is

$$Z_r' = \frac{Z_r}{n^2} = \frac{7.54}{324} \approx \boxed{0.0233\,\Omega}. \tag{28}$$

These are referred equivalent values for analysis only. The physical resonant inductor and capacitor remain on the HV side.

# 10 Simulation Notes

The following parameters can be entered in the Simulink model for the ideal resonant tank:

Table 2: HV-side LC tank parameters

| **Parameter** | **Value** |
|---|---|
| Resonant inductance $L_{cr}$ | $40\,\mu H$ |
| Resonant capacitance $C_{cr}$ | $0.704\,\mu F$ |
| Target resonant frequency $f_r$ | $30\,kHz$ |
| Characteristic impedance $Z_r$ | $7.54\,\Omega$ |

The model should also include the transformer, both active bridges, a suitable load, electrical references, and the required solver configuration.